| INCH-POUND |

MIL-A-28619C(YD1)
26 August 1994
SUPERSEDING
MIL-A-28619B(YD)
19 August 1987

MILITARY SPECIFICATION

ANCHORS, STATO MOORING, 200 TO 15,000 POUNDS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stato mooring anchors, including stabilizers, wedges, and palm extensions.

1.2 Classification. The anchor, with stabilizers, wedges, and palm extensions, shall be one of the following classes, as specified (see 6.2).

Class 1: 200 pounds (lb) nominal weight
Class 2: 3,000 lb nominal weight
Class 3: 6,000 lb nominal weight
Class 4: 9,000 lb nominal weight
Class 5: 12,000 lb nominal weight
Class 6: 15,000 lb nominal weight

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2040

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2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

A-A-1632 - Varnish, Asphalt.

STANDARD

MILITARY

MIL-STD-130 - Identification Marking of U.S. Military Property.

2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)
(Code Ident 07071)

813464 - Budocks Stato Mooring Anchor, 15,000 lb, Hinge for Stabilizers (Alternate Construction).
813503 - Budocks Stato Mooring Anchor, 15,000 lb, General Assembly.
813504 - Budocks Stato Mooring Anchor, 15,000 lb, Details.
813505 - Budocks Stato Mooring Anchor, 15,000 lb, Details.
813507 - Budocks Stato Mooring Anchor, 200 lb, General Assembly.
813508 - Budocks Stato Mooring Anchor, 200 lb, Details.
813509 - Budocks Stato Mooring Anchor, 200 lb, Details.
813510 - Budocks Stato Mooring Anchor, 3,000 lb, General Assembly.
813511 - Budocks Stato Mooring Anchor, 3,000 lb, Details.
813512 - Budocks Stato Mooring Anchor, 3,000 lb, Details.
813513 - Budocks Stato Mooring Anchor, 6,000 lb, General Assembly.
813514 - Budocks Stato Mooring Anchor, 6,000 lb, Details.
813515 - Budocks Stato Mooring Anchor, 6,000 lb, Details.
813516 - Budocks Stato Mooring Anchor, 9,000 lb, General Assembly.
813517 - Budocks Stato Mooring Anchor, 9,000 lb, Details.
813518 - Budocks Stato Mooring Anchor, 9,000 lb, Details.
813519 - Budocks Stato Mooring Anchor, 12,000 lb, General Assembly.
813520 - Budocks Stato Mooring Anchor, 12,000 lb, Details.
813521 - Budocks Stato Mooring Anchor, 12,000 lb, Details.
813561 - Budocks Stato Mooring Anchor, 9,000 lb, Hinge for Stabilizers (Alternate Construction).
813583 - Budocks Stato Mooring Anchor, 6,000 lb, Hinge for Stabilizers (Alternate Construction).
813584 - Budocks Stato Mooring Anchor, 3,000 lb, Hinge for Stabilizers (Alternate Construction).

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879085 - Budocks Stato Mooring Anchor, 12,000 lb, Hinge for Stabilizers (Alternate Construction).

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

ASTM

A36/A36M - Structural Steel.
A275/A275M - Magnetic Particle Examination of Steel Forgings.
A325 - Structural Bolts, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength.
A668/A668M - Steel Forgings, Carbon and Alloy, for General Industrial Use.
D3951 - Commercial Packaging.
E4 - Force Verification of Testing Machines.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Steel.

(Application for copies should be addressed to American Welding Society, Inc., 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135-3030.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SP 10 - Near-White Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4516 Henry Street, Suite 301, Pittsburgh, PA 15213-3728.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specified.

3.3 Design. The anchor shall be designed to have a holding power to weight ratio of 15 to 1 in mud bottoms and 20 to 1 in sand bottoms. Each class of anchor shall be in accordance with the applicable drawings as shown in table I.

TABLE I. Applicable drawings.

Class	Nominal weight (lb)	Applicable Drawing No.
1	200	813507 813508 813509
2	3,000	813510 813511 813512 813584
3	6,000	813513 813514 813515 812483
4	9,000	813516 813517 813518 813561
5	12,000	813519 813520 813521 879085

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TABLE I. Applicable drawings. (continued)

Class	Nominal weight (lb)	Applicable Drawing No.
6	15,000	813503 813504 813505 813464

3.4 Construction. The complete anchor assembly shall conform to the requirements and dimensions specified herein and on the applicable drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. The tolerance criteria of the fluke angle with respect to the anchor shank shall be 50 ±1 degrees (°) for mud use, and 34 ±1° for sand use.

3.4.1 Shank. The shank shall be either cut from structural steel conforming to ASTM A36, or forged from steel conforming to ASTM A668, class D. The finished shank shall conform to the dimension of the applicable drawings.

3.4.2 Stabilizers. Stabilizers for class 1 anchors shall be welded to the anchor as shown on the applicable drawings. Stabilizers for class 2 through class 6 shall be hinged or welded as specified (see 6.2). Welded and hinged stabilizers shall conform to the dimensions shown on the applicable drawings.

3.4.3 Wedges. Unless otherwise specified (see 6.2), wedges shall be provided. When provided, the required number and sizes of bolts, nuts, and washers shall also be provided for complete installation. Bolts, nuts, and washers shall be as shown on the applicable drawings; shall conform to ASTM A325; and shall be heat treated to 120,000 ±5,000 pound-force per square inch tensile strength.

3.4.4 Palm extensions. Unless otherwise specified (see 6.2), palm extensions shall be provided for field welding to the tripping palms. When palm extensions are to be shop welded to the tripping palms, an alternate fabrication method may be used. The alternate method permits the palm extension and the tripping palm to be fabricated from one continuous piece of structural steel. Palm extensions shall conform to the dimensions of the applicable drawings.

3.4.5 Shackle and pin. A shackle and pin conforming to the applicable drawing and to ASTM A668, class K, shall be provided.

3.5 Performance.

3.5.1 Proof load. The complete anchor assembly shall be capable of withstanding a proof load for the applicable class of anchor as shown in table II without damage, permanent deformation, cracks, or distortion. The proof load shall be applied as shown on the applicable drawing.

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TABLE II. Proof loads for anchors by class.

Class	Weight	Proof load
	lb	lb
1	200	10,000
2	3,000	72,600
3	6,000	121,000
4	9,000	156,000
5	12,000	185,000
6	15,000	214,000

3.6 Cleaning and coating. After testing and examination and before coating, all parts shall be cleaned by removal of all oil, grease, loose scale, and other foreign substances which would prevent bonding of the coating to the metal. Anchor parts shall be shot blasted to near white metal SSPC-10, in accordance with the Steel Structures Painting Council. After cleaning, all parts shall receive one coat of asphalt varnish conforming to A-A-1632. The asphalt coating shall be a minimum of 3 mils thick and applied in accordance with manufacturer's recommendations.

3.7 Dimensions. The dimensions and tolerances for each class of anchor and component parts shall be in accordance with the applicable drawings.

3.7.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and finished product, provided form, fit, and function requirements are satisfied.

3.8 Weight. The nominal weight of the assembled anchor shall be as specified in table II for the applicable class of anchor.

3.9 Identification marking. Marking of each anchor shall be in accordance with MIL-STD-130. The following information shall be stamped or raised in letters not less than 3/4-inch high on both the shank and tripping palm of the anchor assembly: Stamped markings shall be round face characters with a minimum of 0.06 inch depth; raised markings shall have a minimum 0.06 relief.

ANCHOR, STATO MOORING

Class: _____
Nominal weight (lb): _____
Holding power to weight ratio: Mud: 15:1 Sand: 20:1
Year fabricated: _____
Manufacturer's name: _____
U.S. NAVY

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3.10 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.11 Workmanship.

3.11.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.11.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.11.3 Welding. Welding procedures shall be in accordance with AWS D1.1 for dynamically loaded structures. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.11.4 Forgings. Forgings shall be of uniform quality free of detrimental flash, scale, cracks, hard spots, colt shuts, or other defects which might render the forging unsound for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Component and material inspections. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection, when specified (see 3.1 and 6.2), shall be performed on one anchor, with stabilizers, wedges, and palm extensions as required. This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory, provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.1.1 Disposition of first article sample. Disposition of first article sample shall be as specified in the contract or purchase order (see 6.2).

4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2), the quality conformance inspection shall consist of the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5.

4.3 Examination. Each anchor, with stabilizers, wedges and palm shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations in accordance with AWS D1.1 and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. Each anchor shall receive the tests of 4.4.1 and 4.4.2. Failure to pass any test shall constitute failure. Repair procedures and repair examinations shall be in accordance with AWS D1.1; procedures must be reviewed and approved by the Government representative prior to repair.

4.4.1 Proof test. Each anchor shall be assembled with shackles, welded or hinged stabilizers, wedges, and palm extensions, and shall be subjected to the applicable proof load as specified in table II. After attachment of the proof-test load, the welds around the stopper, the bearing block, the fluke stiffeners, and the area along the palm-to-fluke joints shall be inspected for permanent deformation, cracks, or distortion. Appearance of any one or combination of the above discrepancies shall be cause for rejection. Anchors shall be gaged before and after the proof test to determine any permanent deformation or distortion caused by the load. Proof test machine shall be certified in accordance with ASTM E4.

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4.4.2 Magnetic particle test. All forged components and all welds shall be magnetic tested. After the proof load test specified in 4.4.1 and prior to cleaning and coating (see 3.6), all forged components of the anchor shall be subjected to the magnetic particle test in accordance with ASTM A275. All welds shall be examined and evaluated in accordance with AWS D1.1 for dynamically loaded structures.

4.5 Packaging inspection. The packing and marking of the anchors shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified in the contract or purchase order (see 6.2), the preservation, packing and marking shall be in accordance with ASTM D3951.

5.1.1 Matchmarking. Unless otherwise specified (see 6.2), parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Stato mooring anchors are for use as permanent mooring type anchors for use in mud or sand bottoms.

6.1.1 Anchor use data. Stato mooring anchor characteristics were developed to obtain a high holding power to weight ratio of 15 to 1 in mud and 20 to 1 in sand. For further information on stato anchors, a summary of the design, tests, and applications are given in the Naval Facilities Engineering Command Design Manual DM-26 for Harbor and Coastal Facilities.

6.1.2 Mud bottom use. Mud bottom use conditions require that no wedge inserts be used. This permits the flukes to rotate and reach the optimum 50° angle between shank and fluke. The tripping palm extensions are required to achieve this within a reasonable drag distance.

6.1.3 Sand bottom use. Sand bottom use conditions require that wedge inserts must be used. This limits the fluke rotation to an optimum 34°. Tripping palm extensions are not required, but do not detract from the anchor's use in sand.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class of anchor required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When stabilizers for class 2 through class 6 are to be hinged or welded (see 3.4.2).

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e. When wedges are not required (see 3.4.3).
f. When palm extensions are not to be provided for field welding (see 3.4.4).
g. When a first article inspection is required (see 4.2.1).
h. Disposition of the first article sample (see 4.2.1.1).
i. Quality conformance inspection, if other than as specified (see 4.2.2).
j. When preservation, packing and marking is other than as specified (see 5.1).
k. When matchmarking is not required (see 5.1.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 227.405-70 are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements. As a minimum the following shall be submitted:

a. Certificate of Compliance
b. Material heat certificates
c. Calibration certificates with name, serial number, and date of calibration of equipment used.
d. Identification of coating manufacturer and brand name.
e. Proof load certificate.
f. Magnetic particle inspection report.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

Palm extensions
Shank
Stabilizers
Wedges

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy - YD1

Preparing Activity:
Navy - YD1

Review Activity:
Navy - SH

(Project 2040-N218)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-28619C(YD1)	940826

3. DOCUMENT TITLE
ANCHORS, STATO MOORING, 200 TO 15,000 POUNDS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
R. J. BRICKEY	805-982-5593	551-5593
c. ADDRESS (Include Zip Code) COMMANDING OFFICER, NCBC CODE 1564B 1000 23RD AVENUE PORT HUENEME, CA 93043-4301	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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